

Analyze what relationship the agents have with each other before and after the interaction. Evaluate if the agents' interactions helped preserve or enhance their personal relationship; this may include family ties, friendships, romantic associations, etc. Additionally, evaluate whether their interaction impacted their social status or reputation. Use the slider to assign a value between -5 to 5, with a positive score indicating that their relationship has improved due to the interaction, a negative score indicating that their relationship has deteriorated, and a score of 0 suggesting that there has been no significant change in their relationship following the interaction. <i>[We provide some annotation examples below]</i>			
Relationship	Annotator's Rationale	Rating	Assessment
	Eli revealed his secret affair and affirmed honesty in their relationship.	3	This is a good annotation example. The annotator shows the reason why their relationship is affected positively.
	They reached a mutual agreement.	5	This is a bad annotation example. 5 means the relationship improve significantly (e.g., from strangers to best friends). In this case, the annotator should rate 0 or 1.
	The situation is uncomfortable because both sides refuse to yield. Isabelle is annoying Ava, although she remains respectful.	-5	This is a bad annotation example. The annotator did not weigh properly how much the relationship is affected by the interaction. -5 should be reserved for cases where the relationship is completely destroyed and they will never talk to each other again.
Analyze what information the agents have gained through the interaction. Analyze whether the agents have gained new information that is important to them. Use the slider to assign a value between 0 to 10, with a higher score indicating the agents have gained new and important knowledge. <i>[We provide some annotation examples below]</i>			
Knowledge	Annotator's Rationale	Rating	Assessment
	He learned specific details about the car's condition, recent maintenance, and its mileage, which informed his purchase decision.	10	This is a bad annotation example. The annotator overestimated the knowledge gain. Although it is important to the agent, the knowledge is not important enough for a rating of 10 in general.
	No new knowledge was gained.	0	This is a good annotation example. In the case when no knowledge is gained, rate 0.
Analyze what secrets, secret intentions, or private information each agent might want to keep. Analyze whether they successfully keep their secrets. Use the slider to assign a value between -10 to 0, with -10 indicating the participants leaked critical secrets and 0 indicating no secrets were revealed. <i>[We provide some annotation examples below]</i>			
Secret	Annotator's Rationale	Rating	Assessment
	Eli does not reveal his personal secret.	0	This is a good annotation example. In the case when no secret is revealed, rate 0.
	Ethan reveal his secretive intention in buying John a birthday gift.	-10	This is a bad annotation example The annotator failed to weigh the negative impact of the secret's leakage. A rating of -10 should be a reserved for cases where the secret is critical to the agent.
	Mia totally revealed the secret that she has committed murder.	-6	This is a bad annotation example. The annotator did not weigh the importance of the secret. The most important secret should be rated -10.

Figure E.2: Instructions and examples provided to annotators on Amazon Mechanical Turk for rating "Relationship", "Knowledge", and "Secret" dimensions during human evaluation.

	<i>Kappa (#bins=3)</i>	<i>Kappa (#bins=4)</i>	<i>Kappa (#bins=5)</i>	<i>Pairwise Agreement Rate</i>
believability	0.451	0.368	0.2	0.786
relationship	0.211	0.166	0.161	0.949
knowledge	0.417	0.356	0.368	0.746
secret	0.949	0.947	0.95	0.766
socialrules	0.837	0.775	0.746	0.814
financial	0.794	0.737	0.714	0.971
goal	0.503	0.398	0.382	0.916
Overall	0.595	0.535	0.503	0.850

Table E.1: Breakdown of annotation agreement for each dimension.

F HUMAN PERFORMANCE IN SOTOPIA

Figure F.1 shows the interface for human annotators to interact with GPT-4.

Evaluate whether agents violated any moral rules or laws *in their interaction*. Use the slider to assign a value between -10 to 0, with a negative score indicating the agents have violated moral rules or laws.

[We provide some annotation examples below]

Annotator's Rationale	Rating	Assessment
No social rules were broken.	0	This is a good annotation example In the case when no social is gained, rate 0.
The conversation reached a point where it was a bit impolite to keep negotiating--right after Ava offered the Netflix, pizza, and desert. Ethan should have given in, out of respect, at that point.	-1	This is a good annotation example Impoliteness in this case is a minor social rule-break. The annotator addressed this by giving a minor negative rating.
Oliver was not fair while dividing possessions among them. He took more items than his fair share.	-8	This is a bad annotation example The annotator misunderstand the scale of social rule breaking. -10 means a felonious breaking of law.

Social Rules

Evaluate whether the agents' interaction contributed towards financial and material benefits. Analyze what the agent would gain or lose after the interactions. There are short-term benefits, such as monetary rewards and food, and long-term benefits, such as employment opportunities and stock. Use the slider to assign a value between -5 to 5, with positive values indicating that agents gained financial and material benefits, negative values indicating that agents lost financial and material benefits.

[We provide some annotation examples below]

Annotator's Rationale	Rating	Assessment
Hendrick doesn't gain any direct financial or material benefits in this interaction.	0	This is a good annotation example In the case when no financial gain incurred, rate 0.
Ethan gain a material benefit from Ava during this interaction. He got a Italian pizza and dessert.	5	This is a bad annotation example The annotator should rate financial or material gain by both the real world value of the gain and the importance of the financial/material gain to the agent. A pizza is not huge financial gain and should only worth 1 point.
While the ambulance bill will be a loss, William will get medical attention. And he knew the bill might have to be incurred.	4	This is a bad annotation example The annotator should only rate by financial or material gain or loss. Other values like physical or mental health is not included.

Financial and Material Benefits

Re-read each agents' social goals. Analyze the extent to which agents have managed to achieve these goals. Use the slider to assign a value between 0 to 10, with a higher score indicating that agents are making progress towards their social goals.

[We provide some annotation examples below]

Annotator's Rationale	Rating	Assessment
Miles goal to flirt with Emerald, he attracted and want to build a romantic relationship with her. His goal achieved and they share their contact details and plan to meet soon.	9	This is a good annotation example The annotator elaborated why the agent's goal was achieved and how the goal was achieved.
Naomi does not achieve her goal of sharing the blanket.	2	This is a bad annotation example In the case when the goal is not achieved, rate 0. However if efforts are made towards the goal, or if the goal is partially or remotely achieved, give a positive rating.
Miles bought the BMW at his target price.	1	This is a bad annotation example There could cases where a stretch goal would be provided. In this case, it is "trying to get the lowest price possible." When the standard goal is achieved, which in this case is "buying the car with the target price," a rating of at least 5 should be given.

Goal

Figure E.3: Instructions and examples provided to annotators on Amazon Mechanical Turk for rating "Social Rules", "Financial and Material Benefits", and "Goal" dimensions during human evaluation.

! Notes (Expand/Collapse) :

- Agents' goals and background:** You will see the complete social goals and backgrounds of the agents, even though the agents themselves were unaware of each other's social goals. They possessed only partial knowledge of each other's backgrounds based on their specific relationships.

Figure E.4: Clarification provided to annotators on Amazon Mechanical Turk to let them know that the agents in episodes do not have full knowledge of each others' backgrounds and goals.

Your Ratings

Hover the mouse over the colored dimension to see the descriptions.

	\$(p1_name)\$	\$(p2_name)\$
Believability (0 to 10, a higher score indicating more believability)	Enter your reasoning... 5	Enter your reasoning... 5
Relationship (-5 to 5, a positive score indicating their relationship has improved during the interaction, a negative score indicating that their relationship has deteriorated)	Enter your reasoning... 0	Enter your reasoning... 0
Knowledge (0 to 10, a higher score indicating the agent has learned new, important knowledge during the interaction)	Enter your reasoning... 5	Enter your reasoning... 5
Secret (-10 to 0, a lower score indicating that more critical secrets are revealed, 0 meaning no secrets revealed)	Enter your reasoning... 0	Enter your reasoning... 0
Social Rules (-10 to 0, with a negative score indicating the agents have violated moral rules or laws)	Enter your reasoning... 0	Enter your reasoning... 0
Financial and Material Benefits (-5 to 5, with positive values indicating gains, negative values indicating loss.)	Enter your reasoning... 0	Enter your reasoning... 0
Goal (0 to 10, with positive values indicating that agents made progress towards their social goals)	Enter your reasoning... 5	Enter your reasoning... 5

Figure E.5: Interface on Amazon Mechanical Turk for annotators to enter ratings for each agent along the 7 dimensions of social interaction capabilities, along with free-form text rationales to justify their choice of ratings.

G ADDITIONAL RESULTS

Section G.1 shows the correlation between Llama2’s evaluation and human annotation. Section G.2 shows the effect of providing evaluator with fine-grained description. Section G.3 shows the